

LOCKOUT TAGOUT TRYOUT

A STUDY GUIDE ON ZERO ENERGY VERIFICATION

INTRODUCTION

The EV Charger job sites present invisible electrical hazards. At Chargeie, our field operations involve deploying high-voltage infrastructure where standard safety isn't enough. Lockout Tagout Tryout (LOTOTO) takes basic protection to a critical final stage. It isn't just about applying a lock; it is about verifying absolute isolation before a single tool touches a live circuit.

In the presence of lethal electrical hazards, absolute isolation is the only guarantee.

WHY IT MATTERS

- EV infrastructure operates on multi-ton electrical currents.
- Unexpected energization during service causes catastrophic, fatal injuries.
- Warning tags alone do not physically stop a circuit from active conduction.

Every technician deserves to return home safely to their family.



WHEN YOU ARE AT RISK

Main Switchgear Work

- Interfacing with commercial panels and electrical rooms.

Pedestal Maintenance

- Servicing charger internals where residual stored energy exists.

Shared Contractor Sites

- Active sites where other trades can accidentally flip a breaker

Complacency Breaks

- Rushing through an installation and skipping the verification step.

When you are on site.

THE COST OF COMPLACENCY

- Severe arc flash explosions and fatal electrocution.
- Irreparable property damage to client facilities and commercial panels.
- Immediate OSHA investigation, heavy corporate penalties, and complete stop-work orders.

One mistake. Massive impact.

THE 9 STEPS OF LOTOTO

1. Prepare for Shutdown

- Identify all energy types—predominantly 480-volt commercial feeds—and map every single isolation point before touching the switchgear. Know what controls the circuit.

2. Notify Affected Personnel

- Communicate directly with facility managers and any shared site contractors. Inform them that the specific chargers or main panel segments are going dark.

3. Equipment Shutdown

- Power down the charging pedestals or infrastructure using standard operating controls. Never pull a disconnect lever while the system is under full operational load.

4. Isolate the Energy Source

- Throw the main service disconnect handle down into the off position. Cut the line connection physically from the incoming grid.

5. Apply Lockout Devices

- Secure a standardized, non-conductive lockout hasp onto the disconnect switch. Snap your personal padlocked red lock onto the hasp. Your lock, your key.

6. Apply Tagout Warning Tags

- Affix a highly visible, legible danger tag to your lock. It must state your name, "Chargie," the date, and the explicit reason for the isolation.

7. Control Stored Energy

- Bleed the remaining lines. Check internal charger pedestals for capacitive residual energy or stored voltage that lingers after the breaker trips.

8. The "Tryout" (Verification)

- **The critical line.** Try to restart the equipment using local controls or tests. Verify the panel or pedestal is completely unresponsive. Test the circuit with a calibrated multi-meter to ensure absolute zero energy.

9. Perform the Service

- Only now—with zero voltage verified—do tools touch the circuit. You are protected by a physical, locked barrier.

THE DE-ISOLATION PROTOCOL

- The Clearout: Inspect the inside of the switchgear and pedestal. Verify all tools, loose wires, and testing leads are physically removed from the terminal cabinet.
- The Shielding: Reinstall all internal protective barriers, dead-front panels, and equipment covers. No live components left exposed.
- The Accountability: Ensure all other technicians working the shared site have cleared the machine and removed their personal locks from the hasp first.
- The Lock Stripping: Personally remove your own padlock and danger tag from the isolation point. Never let another tech snap your lock off.
- The Restoration: Re-engage the main service disconnect handle back into the ON position to re-energize the infrastructure.
- The Diagnostic: Run a live terminal voltage check to confirm the EV chargers are pulling clean, balanced power and reporting online

MOVE FAST, EXECUTE CLEAN

SECONDS OF THINKING PREVENTS WEEKS OF HEALING.